

ST LAWRENCE, Australia

WMO: 953690

Lat: 22.3472S

Lon: 149.5242E

Elev: 10

StdP: 101.21

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	9.3	-2.4	3.1	17.6	-0.3	3.7	16.6	7.3	19.7	6.7	19.5	1.0	180	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.8	33.2	24.3	32.3	24.1	31.6	23.8	26.7	30.5	26.1	29.8	25.6	29.2	4.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.7	21.0	28.7	25.1	20.3	28.3	24.6	19.6	27.9	84.0	30.6	81.4	29.9	79.0	29.2	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.8	6.2	DB	4.8	36.5	1.5	1.8	3.7	37.8	2.8	38.8	2.0	39.8	0.9	41.1	(4)
(5)			WB	3.0	28.0	1.4	1.0	1.9	28.7	1.1	29.3	0.3	29.9	-0.8	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	27.2	27.0	26.1	23.8	20.8	18.4	17.6	18.3	20.9	23.4	25.2	26.7	(6)
(7)		DBStd	3.93	1.47	1.56	1.45	1.63	2.13	2.39	2.22	2.07	1.94	1.89	1.61	1.64	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	102	0	0	0	0	6	29	40	25	2	0	0	0	(9)
(10)		CDD10.0	4713	532	476	498	414	336	251	236	258	326	414	455	518	(10)
(11)		CDD18.3	1773	274	242	240	164	83	29	17	24	78	156	205	259	(11)
(12)		CDH23.3	15113	2832	2404	2119	1093	418	107	73	144	502	1099	1721	2601	(12)
(13)	CDH26.7	4659	1049	876	701	261	41	3	2	4	61	223	502	937	(13)	
(14)	Wind	WSAvg	2.7	3.0	2.7	2.7	2.4	2.2	2.1	2.3	2.7	3.2	3.3	3.2	(14)	
(15)	Precipitation	PrecAvg	990	170	169	129	54	58	31	25	28	22	44	92	148	(15)
(16)		PrecMax	1733	643	679	513	221	299	265	121	219	140	133	428	613	(16)
(17)		PrecMin	381	0	12	1	2	0	0	0	0	0	0	0	1	(17)
(18)		PrecStd	357	150	167	134	55	60	41	30	42	34	41	85	134	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	34.2	33.5	31.4	29.0	26.9	26.4	27.1	30.4	31.7	33.5	34.5	(19)
(20)			MCWB	25.5	24.9	24.1	22.3	20.1	19.4	17.5	17.5	18.8	21.1	20.7	24.3	(20)
(21)		2%	DB	32.7	32.9	32.1	30.2	27.8	25.7	25.0	25.7	28.2	29.7	31.1	32.5	(21)
(22)			MCWB	25.1	24.4	23.6	21.6	19.8	18.5	17.4	17.6	18.9	21.9	22.4	24.4	(22)
(23)		5%	DB	31.9	32.0	31.1	29.2	26.9	24.6	24.1	24.9	27.0	28.7	30.2	31.5	(23)
(24)			MCWB	24.5	24.4	23.5	21.4	19.3	18.1	16.9	17.5	19.3	21.4	22.3	23.9	(24)
(25)		10%	DB	31.0	31.0	30.2	28.3	25.8	23.5	23.1	24.0	26.1	27.9	29.4	30.7	(25)
(26)			MCWB	24.0	24.2	23.2	21.2	18.9	17.4	16.4	16.9	18.9	20.9	22.1	23.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	27.3	27.0	25.3	23.3	22.0	21.0	21.3	23.0	24.7	25.7	27.2	(27)
(28)			MCDWB	31.8	30.8	30.5	28.5	26.2	24.0	23.6	24.2	26.8	28.5	29.9	30.8	(28)
(29)		2%	WB	26.6	26.5	25.9	24.1	22.0	20.7	19.9	20.3	22.0	23.8	24.7	26.3	(29)
(30)			MCDWB	30.7	30.1	29.1	27.4	25.4	23.4	22.6	23.4	25.4	27.3	28.5	30.2	(30)
(31)		5%	WB	26.0	25.9	25.2	23.2	21.1	19.7	18.9	19.5	21.2	23.1	24.0	25.5	(31)
(32)			MCDWB	29.7	29.5	28.6	26.7	24.7	22.8	21.9	22.7	24.8	26.7	28.0	29.4	(32)
(33)		10%	WB	25.4	25.4	24.5	22.4	20.3	18.9	17.9	18.5	20.5	22.3	23.3	24.9	(33)
(34)			MCDWB	29.1	29.0	28.1	26.2	24.0	22.2	21.3	22.0	24.1	26.1	27.4	28.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	7.8	8.4	9.3	10.4	10.5	11.6	12.0	11.1	9.5	8.6	8.1	(35)
(36)			MCDBR	8.9	9.3	9.7	10.5	11.0	11.1	12.1	12.5	11.9	10.0	9.4	8.9	(36)
(37)			MCWBR	3.0	3.0	3.1	3.7	4.6	5.3	5.7	5.9	5.1	3.9	3.4	3.0	(37)
(38)		5% WB	MCDWB	7.6	7.5	7.5	8.1	9.0	8.8	9.0	9.5	9.1	8.2	7.9	7.7	(38)
(39)			MCWBR	3.2	3.1	3.2	3.5	4.1	4.9	4.9	5.0	4.3	3.7	3.3	3.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.404	0.392	0.373	0.354	0.335	0.323	0.313	0.324	0.357	0.386	0.405	0.403	(40)	
(41)		taud	2.482	2.519	2.560	2.591	2.588	2.620	2.626	2.581	2.486	2.435	2.414	2.457	(41)	
(42)		Ebn at Noon	941	940	931	909	888	881	904	926	933	934	934	943	(42)	
(43)		Edh at Noon	118	111	103	92	85	79	81	91	109	120	125	121	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.40	6.05	5.55	4.87	4.20	3.68	4.15	5.03	6.08	6.73	6.90	6.86	(44)	
(45)		RadStd	0.46	0.63	0.48	0.49	0.24	0.33	0.31	0.34	0.44	0.41	0.79	0.70	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	-0.47	-0.42	-0.45	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page